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Sound Insulation: Key Concepts and Technologies

Jonty Mago, Sunali, Ashutosh Negi, J. Stuart Bolton, and S. Fatima

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Abstract

The modern era of urbanization and technological advancements has witnessed a substantial increase in noise levels, making it paramount to understand and innovate in the field of sound insulation. This chapter explores the critical and continually evolving field of sound insulation, addressing major public health concerns associated with noise pollution. It highlights the prevalence of noise pollution in urban environments and its diverse detrimental health effects, such as hearing loss, stress, cardiovascular diseases, and cognitive impairment. Consequently, the discussion delves into noise control strategies, distinguishing between active and passive noise control mechanisms. Active noise control involves using electronic means to cancel out sound waves, while passive noise control relies on materials and designs to absorb or reflect sound energy. Moreover, a significant section focuses on sound insulation materials, which are essential for passive noise control, as they reflect sound waves. It also elaborates on these materials by detailing their performance metrics, typical sound transmission loss (STL) curve, factors affecting STL, and their effectiveness in mitigating noise pollution. Additionally, it discusses experimental methods for measuring STL and predictive models for assessing STL in materials. A comprehensive review of commercially available sound insulation materials introduces traditional materials alongside recent advancements such as metamaterials, perforated panels, and polymer composites. Furthermore, the potential of using recycled or waste-derived materials in sound insulation is thoroughly discussed, spotlighting their practical applications and future research prospects. In conclusion, the chapter reviews sound insulators and their integral role in noise control strategies, thereby contributing to improvements in public health. Through a detailed review of materials, strategies, and experimental methods, this chapter offers valuable insights into the current state and future directions of sound insulation technology.

Keywords

Acoustic materials · Experimental methods · Noise control strategies · Noise pollution · Performance metrics · Practical applications · Sound transmission loss · Waste and recycled materials

Introduction

In the contemporary world, noise pollution has emerged as a modern plague, posing considerable environmental and public health concerns (Goines and Hagler 2007). The relentless noise from buildings, vehicles, ships, aircraft, and machines disrupts people's daily lives and leads to various health issues. Studies have shown that noise pollution can cause speech interference and sleep disturbances, provoke annoyance, and lead to hearing loss in severe cases (Han et al. 2015; Swinburn et al. 2015). Urban areas in countries like India, particularly cities such as Delhi and Mumbai, are among the worst affected globally in terms of noise pollution levels (Mongabay 2020).

The World Health Organization (WHO) has highlighted the risk of hearing loss from noise exposure, which is estimated to affect 1 billion people aged 12–35 worldwide, based on their 2017 report (WHO and ITU 2019). The adverse impact of noise pollution extends beyond hearing loss, encompassing mental stress, cardiovascular diseases, hypertension, and cognitive impairment. Occupational exposure to loud environments can lead to a range of physiological difficulties, including but not limited to headaches, heart problems, high blood pressure, reduced productivity, and an increased risk of workplace accidents (Smith 1989; Munjal 2013). The chronic nature of noise pollution amplifies its impact, resulting in long-term public health complications. Interestingly, the COVID-19 pandemic, while temporarily reducing human activity and noise pollution, underscored the importance of understanding its long-term implications and developing more effective management strategies (Caraka et al. 2021; Gheibi et al. 2022).

Efforts to control noise pollution have evolved to focus on modifications at the source, path, and receiver levels (Mohanty and Fatima 2013; NPTEL 2023). A fundamental aspect of these strategies is distinguishing between active and passive noise control. Active noise control involves using an additional secondary sound source, such as noise-canceling headphones, to negate the effect of the present sound. On the other hand, passive noise control, often more economical and energy efficient as it does not require external power, employs materials that absorb or reflect sound energy, referred to as acoustic materials (Rajappan et al. 2017).

Although modifying the noise source is the most effective and cost-efficient method, it is not always feasible, particularly with pre-installed machinery. In such cases, attention shifts to path and receiver modifications, vital for noise control despite being more challenging and costly. The extensive application of acoustic materials, especially in path modification through insulators/barriers, absorbers, and damping materials, plays a pivotal role in these strategies (Reynolds 1981; Bies and

Hansen 2017; NPTEL 2023). For a comprehensive understanding of noise control, Fig. 1 illustrates the Source-Path-Receiver model, which encapsulates the strategies employed in mitigating noise pollution by targeting its source, the path it travels, and the receiver. Furthermore, Table 1 provides a detailed list of diverse strategies and techniques applicable at different levels in noise control, drawn from extensive literature, offering insights for acoustical engineers and noise control professionals. These strategies range from choosing the quietest machine source available to constructing personnel enclosures, installing enclosures, using absorbing materials, and providing earplugs or earmuffs for personnel.

Acoustic materials, integral to noise control, have markedly expanded their reach due to widespread application across various sectors (Bansod et al. 2016; Kumar and

Table 1 Strategies for noise control at different levels (Crocker 2007; Munjal 2013)

Source	Path	Receiver
Choose the quietest machine source available	Use barriers	Construct personnel enclosures
Reduce force amplitude or apply force slowly	Install enclosures	Rotate personnel to reduce exposure time
Balancing moving parts	Use absorbing materials	Locate personal remotely from the source
Improve bearing alignment	Mismatch impedance of materials	Provide earplugs or earmuffs for personnel
Use better lubrication	Use damping materials	–
Use dynamic absorbers	Install vibration isolators	–
Use softer material for impacting the surface	Use lined ducts and plenum chambers	–
Change the natural frequency of the machine elements	Use flexible ductwork	–
Reduce radiating surface area by adding holes	Use vibration breaks in duct work	–
Stagger time of machine operations in the plant	Install reactive or dissipative mufflers	–

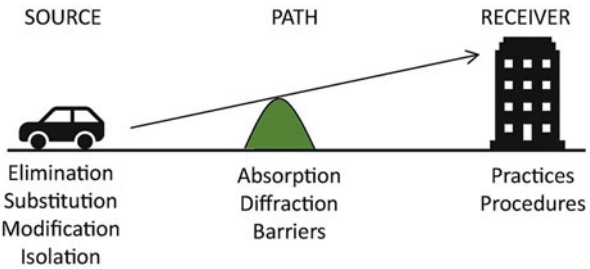


Fig. 1 The Source-Path-Receiver model of noise control: This graphical representation illustrates a comprehensive approach to mitigating noise pollution, focusing on the noise source, the path of transmission, and the receiver's point (Larsen et al. 2022). (Reproduced under the CC BY 4.0 license)

Lee 2019). The global acoustic insulation market, valued at $14.1 billion in 2021, is anticipated to grow to $17.1 billion by 2026, with a Compound Annual Growth Rate (CAGR) of 3.9% (Markets and Markets 2021). This growth is propelled by the rising demand across various sectors, stringent environmental regulations, and growing awareness of the health implications of noise pollution. Acoustic materials are primarily classified into two types: absorbers and insulators/barriers. Absorbers are designed to absorb sound energy, while insulators or barriers are used to block or reflect sound waves. The effectiveness of these materials is measured using various performance metrics such as Sound Absorption Coefficient (SAC), Noise Reduction Coefficient (NRC), Insertion Loss (IL), Noise Reduction (NR), Sound Transmission Loss (STL), and Sound Transmission Class (STC). These metrics are essential for determining the suitability of acoustic materials for specific applications, ensuring that noise control measures are both practical and efficient (Crocker 2007; Mohanty and Fatima 2013).

Focusing specifically on insulators, this category is pivotal in passive noise control strategies, primarily functioning by blocking or reflecting sound energy. The STL metric is a critical indicator of an insulator's effectiveness, measuring its ability to prevent sound transmission from one area to another. This chapter delves into a comprehensive examination of sound insulation materials, exploring their performance metrics, factors affecting the STL of materials, and the typical STL curve. It includes an overview of experimental methods for measuring STL, such as those outlined by the American Society for Testing and Materials (ASTM) and International Organization for Standardization (ISO) standards. Additionally, it discusses models for predicting STL in various materials.

Moreover, the chapter reviews commercially available sound insulation materials, highlighting recent advances in the field, such as developing metamaterials, perforated panels, and polymer composites. It also considers the potential of recycled or waste-derived materials for sound insulation, reflecting a growing interest in sustainable and environmentally friendly solutions. The practical applications of these materials are varied, underscoring their importance in a wide range of noise control scenarios, from urban planning and construction to manufacturing and workplace environments. In conclusion, the chapter summarizes future directions in sound insulation material research, emphasizing the ongoing need for innovative approaches to managing noise pollution and its impact on public health.

Acoustic Materials and Their Performance Metrics

Categories of Acoustic Materials

Acoustic or noise control materials are engineered to mitigate noise by managing sound waves through direction, manipulation, and containment. The effectiveness of these materials in noise control depends on their ability to effectively manage sound wave reflection, dissipation, and transmission upon impact. Based on the desired outcome of noise reduction, the method of sound energy management, and their

interaction with sound waves, these materials are broadly categorized into two groups: Absorbers and Insulators/Barriers (Harris 1991; Mohanty 2002; Kim and Bolton 2017).

(a) **Absorber (Sound Absorbing):** Absorber materials are specially designed to reduce noise by absorbing sound waves and converting them into heat, lowering sound levels (Kuntz 2008; Raj et al. 2020). These materials primarily control noise within the environment where they are installed (referred to as medium 1) and, to a lesser extent, in the adjacent environment (medium 2). They are characterized by low sound reflection and transmission capabilities alongside high sound energy absorption capacity. This mechanism is depicted in Fig. 2a, where a red arrow signifies significant energy dissipation. Additionally, absorber materials exhibit minimal sound reflection and transmission, as indicated by the small yellow and blue arrows. Typically, these materials possess a porous or fibrous structure that facilitates the trapping and dissipating of sound waves (Ackermann et al. 1988; Kim et al. 2015).

(b) **Insulator/Barrier (Sound Blocking or Reflecting):** Insulator/Barrier is designed to control noise in an adjacent medium by obstructing sound waves (Xu et al. 2017). It is optimized to reflect a substantial amount of sound, internally dissipate more sound energy, and transmit the least amount of sound, effectively controlling noise in the secondary medium (medium 2). Figure 2b visually represents their function, with a large yellow arrow showing most sound energy being reflected in the initial medium (medium 1). Furthermore, a significant red arrow illustrates that these materials can convert a portion of sound energy into heat within themselves, thereby reducing sound transmission. The small blue arrow indicates minimal sound transmission through the barrier. Physically, these materials are dense and solid, typically made from heavy, non-porous substances like mass-loaded vinyl or dense foam. Their solidity and density are crucial for their effectiveness in reflecting and dissipating sound, making them suitable for applications requiring significant noise reduction (Wang et al. 2011; Xu et al. 2017).

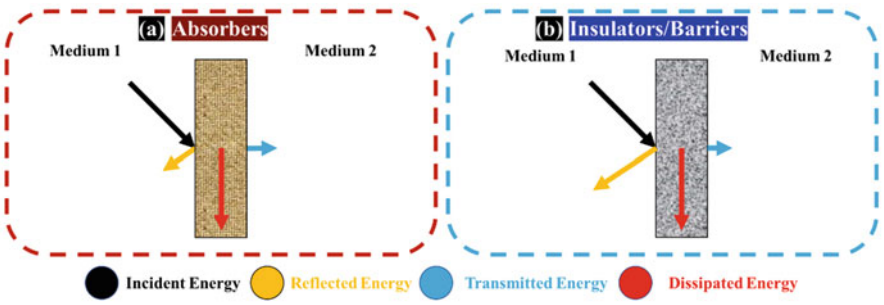


Fig. 2 Acoustic materials are categorized into (**a**) absorbers, reducing noise by converting sound energy to heat with minimal reflection and transmission, and (**b**) barriers, reflecting sound and minimizing transmission for effective noise control

Performance Metrics

In acoustic engineering and design, the effectiveness of acoustic materials is quantified through various performance metrics. Each metric offers a unique perspective on how materials interact with sound, providing crucial insights for their application in diverse acoustic environments. The following section explores the key metrics used to evaluate acoustic materials, discussing their significance, calculation methods, and practical implications in real-world scenarios (Yang et al. 2019; Hongisto et al. 2023).

i. **Noise Reduction (NR):** A fundamental metric in acoustic engineering that quantifies a material's ability to decrease sound levels. It measures the difference in sound pressure levels (SPLs) before and after the sound interacts with a material, assessing its sound attenuation capacity directly. This metric is particularly valuable in environments where reducing ambient noise is essential, such as urban planning or industrial noise control. NR is calculated by subtracting the SPL after the sound passes through the material (SPL_2) from the level before encountering the material (SPL_1), expressed as Eq. 1. This straightforward calculation makes NR a widely used tool in evaluating the efficiency of various sound insulation materials for a wide range of applications (Romeu et al. 2002; Zera and Mlynski 2007; McKinley et al. 2011).

$$\mathbf{NR}(\mathrm{dB}) = \mathrm{SPL}_1 - \mathrm{SPL}_2 \tag{1}$$

ii. **Insertion Loss (IL):** IL is a critical performance metric in acoustic materials, highlighting the material's role in specific acoustic environments. It is defined as the difference in SPL at a particular point, measured with and without the presence of the acoustic material. This measurement precisely indicates the material's effectiveness in real-world installations, considering the actual conditions and placement of the material. IL is calculated using Eq. 2, where SPL_1 is the SPL without the material and SPL_2 with the material in place. This metric is fundamental across various applications, from architectural acoustics to environmental noise control, providing valuable insights into the impact of acoustic interventions in urban spaces (Prasad and Crocker 1983; Cobo et al. 1998; Hull 2005).

$$\mathbf{IL}(\mathrm{dB}) = \mathrm{SPL}_1 - \mathrm{SPL}_2 \tag{2}$$

iii. **Sound Transmission Loss (STL):** STL is a comprehensive and crucial metric in evaluating acoustic materials, particularly relevant in contexts where preventing sound transmission is paramount. STL denotes the difference between the incident and transmitted sound intensity levels when sound waves encounter a material. This metric signifies the material's ability to block sound transmission and plays a pivotal role in designing soundproofing solutions across various industries, including construction and automotive

manufacturing (Varanasi et al. 2013; Kim et al. 2013). The calculation of STL involves determining the intensity transmission coefficient (τ) and is calculated with Eq. 3.

$$\mathbf{TL}\ (\mathrm{dB}) = 10\ \log_{10}\left(\frac{1}{\tau}\right) \tag{3}$$

where τ represents the ratio of transmitted sound intensity ($I_{\text{transmitted}}$) to the incident sound intensity (I_{incident}), underscoring STL's significance through its standard measurement approach, typically executed using a four-microphone impedance tube (Fig. 3) as per (ASTM E2611-19). This method is dedicated to material testing in a controlled environment, while (ASTM E90-09) complements it for diffused field applications. The latter involves testing in a more realistic room-to-room scenario, illustrating how sound transmits through building partitions. While ASTM E2611-19 targets material-level testing, ASTM E90-09 evaluates the performance of partitions within architectural spaces. Such rigorous assessments ensure that materials meet the required standards to effectively reduce sound transmission in diverse applications.

iv. **Sound Absorption Coefficient (SAC):** SAC is a pivotal metric for evaluating the acoustic properties of materials, especially in environments where sound quality and clarity are paramount. SAC quantifies the ratio of incident sound energy that a material absorbs instead of reflecting it into the environment. This property is vital in spaces such as theaters, recording studios, and concert halls, where key objectives are controlling reverberation and improving sound clarity (Sakthivel et al. 2021; Tang and Chuang 2022). The SAC is determined using Eq. 4.



Fig. 3 Illustration of a four-microphone impedance tube (Model: Type 4206-T; Manufacturer: Brüel & Kjær) for measuring STL. This setup features two modular assembly options to accommodate a wide range of frequencies: (**a**) the large tube, with a diameter of 100 mm, for low-frequency measurements (50 Hz to 1.6 kHz), and (**b**) a small tube, with a diameter of 29 mm, for high-frequency analysis (500 Hz to 6.4 kHz) (Hottinger Brüel & Kjær). (With permission of HBK © Hottinger Brüel & Kjær. All rights reserved)

$$\boldsymbol{\alpha}\,(f) = \frac{I_{\text{Absorbed}}}{I_{\text{Incident}}} = 1 - |R|^2 \tag{4}$$

where R denotes the material's reflection coefficient, and the equation relates absorbed (I_{absorbed}) and incident (I_{incident}) sound intensities. This formula highlights the material's capacity to absorb sound across various frequencies. The accuracy of SAC measurements is typically ensured by using a two-microphone impedance tube, in accordance with (ASTM E1050-19), which is ideally suited for controlled, standing wave environment measurements. In contrast, (ASTM C423-22) measures sound absorption in a more generalized, diffused field setting, simulating random incident sound waves. While ASTM E1050-19 focuses on detailed material characterization in a controlled environment, ASTM C423-22 evaluates material performance in conditions that more closely mimic real-world scenarios. The SAC metric, assessed under the specific conditions of ASTM E1050-19 or the diffused field conditions of ASTM C423-22, plays a crucial role in selecting and designing acoustic materials, ensuring they meet the diverse sound absorption requirements of various settings.

v. **Sound Transmission Class (STC):** STC is an essential metric for gaging the effectiveness of soundproofing materials in building construction. It is an integer rating that evaluates the material's ability to reduce sound transmission, with laboratory measurements conducted according to the (ASTM E413-22) standard. The STC rating provides a comparative noise control measure, indicating that loud speech is not heard through walls at higher ratings. Conversely, lower ratings, such as 25, signify poor sound insulation, underscoring the importance of STC in analyzing the sound insulation level of different constructions. This makes it a crucial tool for architects and builders. The efficacy of STC ratings in noise control and the audibility of speech through walls are delineated in Table 2. This table shows that higher STC ratings correspond to better sound insulation, with ratings of 55 and above indicating that normal speech cannot be easily heard through walls. Understanding these STC ratings and their practical implications is vital for ensuring that constructions meet the specific acoustic requirements of their intended use, thereby fostering comfortable and functionally sound environments (Quirt 1986; Byrick 2010).

Table 2 Correlation between STC ratings and acoustic performance—understanding noise control and speech audibility (NGC Testing Services 2023)

STC rating	Noise control level	Speech heard through the wall
55 and more	Excellent	Loud speech not heard
50	Very good	Loud speech, barely audible
40	Good	Loud speech audible as a murmur but unintelligible
30	Marginal	Loud speech understandable
25	Poor	Normal speech understandable

vi. **Noise Reduction Coefficient (NRC):** NRC is a critical metric for quantifying a material's average sound absorption capacity across relevant frequencies. It provides a quick and approximate assessment of a material's sound-absorbing effectiveness (Tang and Yan 2017), especially in general architectural applications. NRC is calculated as the one-third octave average of the SACs at four specific frequencies: 250 Hz, 500 Hz, 1000 Hz, and 2000 Hz. Eq. 5 outlines the NRC calculation:

$$\mathbf{NRC} = \frac{(\alpha_{250} + \alpha_{500} + \alpha_{1000} + \alpha_{2000})}{4} \tag{5}$$

This average is typically rounded to the nearest multiple of 0.05 for ease of use. While NRC offers a convenient single-value metric for comparing different materials, it is important to acknowledge that it may not accurately represent a material's performance at very low or very high frequencies. Despite this limitation, NRC remains a widely used and valuable tool in the initial stages of material selection for acoustic treatments. It assists designers and engineers in making informed decisions regarding the suitability of materials for various sound-absorbing applications (Liu et al. 2013; Lenz 2022).

Typical Sound Transmission Loss Curve

STL is essential for assessing a material's effectiveness in sound blocking and is crucial for architectural acoustics and noise control. It aids in creating quieter environments by illustrating how barriers like walls or windows attenuate sound across frequencies. The typical STL curve, as depicted in Fig. 4, visually depicts this attenuation, showing the relationship between sound frequencies and material properties for an oblique angle of incident sound waves. This curve is divided into regions, each corresponding to a frequency range influenced by the material's unique physical properties, highlighting how various materials contribute to sound attenuation efficiency. The detailed exploration of these regions demonstrates the specific ways in which material's interact with sound frequencies, as further discussed below:

Stiffness Region

The STL curve starts with the stiffness region. In this region, a material's sound-blocking ability is influenced by its rigidity. Edge-constrained stiff barriers effectively block low-frequency sounds with long wavelengths, leading to a higher STL at lower frequencies. This part of the curve shows a 6 dB decrease in STL for every frequency doubling (Wang et al. 2011).

Resonance Region

Ascending in frequency, the curve approaches the resonance region, marked by variable STL values with pronounced peaks and troughs. These variations correspond to the frequencies at which the material or structure's natural vibrations align with the sound waves, potentially enhancing sound transmission (Tocci 1990). Figure 4 also illustrates the influence of damping in this region: without damping, peaks and valleys are noticeable; with low damping, fluctuations lessen; and with high damping, the curve smooths, indicating more effective sound reduction. This emphasizes the importance of damping in reducing STL fluctuations and improving sound insulation. Materials with superior damping properties are shown to be more effective at reducing sound transmission, underscoring the critical relationship between the resonant frequency and damping effects in enhancing sound insulation (Dijckmans et al. 2010).

Mass Law Region

The underlying principle in the mass law region is that heavier barriers are more effective at blocking sound, a property that becomes increasingly pronounced at higher frequencies. This is reflected in the STL curve (Fig. 4), which shows a steady increase of about 6 dB for each frequency doubling, highlighting the significance of a material's mass in sound insulation (Crocker and Price 1969). This principle is quantitatively expressed by Eq. 6.

$$\mathbf{TL_{mass}} = 20\log_{10}(f) + 20\log_{10}(\mathrm{m}) - 47.4 \tag{6}$$

where f is the frequency and m is the mass per unit area of the barrier. This equation underscores the critical influence of material mass on its sound-blocking capabilities, with the linear increase in STL reinforcing that heavier barriers are particularly effective at higher frequencies.

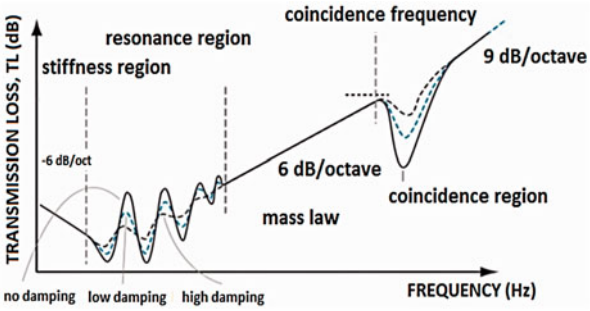


Fig. 4 Typical STL curve across frequency ranges for an oblique angle of incident sound waves (De Rosa et al. 2014) (Pre-approved permission under STM guidelines) © [SAGE Publications]

Coincidence Region

In the coincidence region, a noticeable dip in the STL curve occurs at a frequency where the sound wave's wavelength is the same size as the flexural wavelength in the barrier, causing increased sound penetration. This drop in STL is referred to as the "critical" or "coincident" frequency (f_c), which marks the point where the barrier's vibration resonates with the flexural wave, significantly lowering the material's sound-blocking capability (Peng et al. 2022). The situation is governed by Eq. 7 for the critical or coincident frequency:

$$f_c = \frac{c^2}{2\pi \sin^2(\theta)} \sqrt{\frac{\sigma}{D}} \tag{7}$$

where D is the flexural rigidity, σ the mass per unit area, θ the angle of incidence of the sound wave, and c is the speed of sound. This equation illustrates the interaction between structural factors and the escalation of sound transmission at the frequency where the structural and flexural wavelengths coincide.

Depending on these relationships, this alignment results in a pronounced decline in STL, often quantified at 9 dB per octave. The reduction rate of STL is influenced by the product of flexural stiffness and the combined effects of the angle of incidence and sound speed. This exemplifies the complex interplay of factors that can intensify sound transmission through the perceived protective barrier.

Factors Affecting the Sound Transmission Loss of Materials

Grasping the factors affecting the STL of materials is essential for enhancing their sound insulation properties across various applications—including building construction, the automotive industry, and noise control systems. A comprehensive understanding of these elements facilitates selecting and designing materials tailored for sound insulation in these diverse contexts and optimizes their performance to meet specific requirements (Rabbi et al. 2013; Wareing et al. 2015; Garg et al. 2015; Natsuki and Natsuki 2019; Thanh and Thinh 2019; Söyek Abay and Tanaçan 2022). The key factors that influence material's STL and their sound insulation effectiveness are:

i. **Density and Mass:** Heavier and denser materials generally have higher STL. This is attributed to the "mass law" principle, which states that increasing the mass of a barrier increases its ability to block sound. However, after a certain point, additional mass yields diminishing returns.
ii. **Thickness:** The thickness of a material having a specified density is directly proportional to its STL, especially for lower frequencies. Thicker barriers can vibrate less easily and, therefore, transmit less sound.

iii. **Stiffness and Elastic Properties:** Stiffer materials can transmit sound more efficiently in the coincidence region, leading to lower STL.
iv. **Internal Damping:** Materials with higher internal damping can convert more sound energy into heat, thus reducing sound transmission. Materials like rubber, foam, and fiberglass have high-damping properties.
v. **Resonance:** Every material has a resonant frequency at which it vibrates most efficiently. The material can amplify sound at this frequency, reducing its STL. Designing materials to avoid resonance at common sound frequencies is essential.
vi. **Temperature and Humidity:** These environmental factors can change the physical properties of materials, affecting their STL. For example, higher humidity can affect sound propagation by changing the speed at which sound travels through the air.
vii. **Frequency of Sound:** Materials exhibit varying STL characteristics across different frequencies. Materials may be very effective at blocking high-frequency sounds but less effective at low frequencies, and vice versa.
viii. **Installation and Workmanship:** How a material is installed can significantly impact its sound transmission properties. Improper installation, characterized by gaps, holes, or poor seals, can significantly undermine the STL of a barrier.
ix. **Composite Structures:** Using composite materials (e.g., layers of different materials) can enhance STL, as each layer can be engineered to target specific frequencies or sound types for enhanced sound insulation.
x. **Boundary Conditions and Flanking Paths:** The effectiveness of a material in blocking sound can be affected by how it interacts with other structural elements. Sound can bypass a soundproof material through "Flanking paths," which refers to indirect routes for sound transmissions, such as through ceilings, floors, and adjoining walls.

Understanding and optimizing these factors can develop more efficient materials for sound insulation in various fields. The choice of material and its properties must be aligned with the specific noise control requirements of the application.

Experimental Methods for Measuring Sound Transmission Loss

Adherence to established standards in acoustics ensures uniformity, safety, and performance across various building elements. The ASTM and ISO are paramount among the numerous organizations that set these benchmarks. Their standards cover a wide range of criteria, from STL to the acoustic properties of materials, providing essential guidelines for architects, engineers, and construction professionals. This section explores several essential ASTM and ISO standards, shedding light on their purposes, methodologies, and applicability to the acoustic evaluation and design of building partitions, facades, and other structural components. By adhering to these standards, the industry ensures that buildings meet legal and safety requirements while providing occupants comfort and privacy.

American Society for Testing and Materials Standards

ASTM E90-09

The (ASTM E90-09) standard is a test method for measuring the airborne STL of building partitions such as walls, operable partitions, floor-ceiling assemblies, doors, windows, roofs, panels, and other space-dividing elements in a laboratory setting. It ensures that the partition being tested is the primary sound transmission path between two test rooms designed to create approximately diffuse sound fields. The standard outlines the necessary laboratory conditions, equipment, and procedures for accurate measurement, including specific room conditions, specimen mounting, sound source characteristics, and data collection methods. The results provide a general performance measure for the sound-insulating properties of the tested partitions or elements, applicable to various sound fields they might encounter in actual use.

ASTM E336-20

(ASTM E336-20) is a standard for measuring airborne sound attenuation between rooms in buildings. It prescribes methods for assessing sound isolation, including direct and flanking transmission paths. The standard outlines procedures for measuring NR and Noise Isolation Class (NIC), as well as Normalized Noise Reduction (NNR) and Normalized Noise Isolation Class (NNIC). These measurements are critical for evaluating the acoustic performance of spaces such as residential or office areas, emphasizing specific procedures and metrics for different room volumes and conditions to ensure accurate and relevant results.

ASTM E413-22

(ASTM E413-22) describes the procedure for calculating single-number acoustic ratings from sound attenuation measurements across one-third octave bands in both laboratory and field settings. It applies to various testing methods, including E90, E336, E596, and E1414, assigning specific single-number ratings such as STC, NIC, and Ceiling Attenuation Class (CAC). These classifications are crucial for assessing the sound insulation of building elements, offering a simplified yet comprehensive depiction of their acoustic performance.

ASTM E966-18a

(ASTM E966-18a) provides a comprehensive guide for field measurements of airborne sound attenuation of building facades and facade elements. This standard details methods for evaluating the sound isolation of a building from outdoor noise and assessing sound transmission through building facades, including windows or doors. It offers various measurement techniques suitable for different field situations, highlighting the importance of outdoor sound field geometry and sound incidence angles in calculations. This standard integrates with others for a holistic approach to sound isolation assessment.

ASTM E1123-86

(ASTM E1123-86) outlines practices for mounting test specimens for STL testing of naval and marine ship bulkhead treatment materials. This standard specifies structural requirements and mounting methods for single-layer or composite treatments in ship applications, aiming to accurately measure STL using a specific mounting structure and methods compatible with E90. This approach ensures that the acoustic properties of materials are correctly evaluated for effectiveness in naval and marine settings.

ASTM E1332-22

(ASTM E1332-22) is a standard that classifies the outdoor-indoor sound attenuation of buildings and facade elements. It details methods for calculating single-number ratings that evaluate the isolation from outdoor sound provided by a building, including walls, doors, windows, and the complete structure. These ratings correlate with subjective impressions of a building's ability to mitigate outdoor transportation noise. The standard specifies data collection using one-third octave bands from 80 to 4000 Hz, including definitions for the Outdoor-Indoor Transmission Class (OITC) and the Outdoor-Indoor Noise Isolation Class (OINIC). It is vital for assessing the performance of building elements against specific outdoor sound spectra.

ASTM E1414/E1414M-21a

(ASTM E1414/E1414M-21a) tests airborne sound attenuation between rooms with a common ceiling plenum. It provides a methodology for measuring the sound attenuation of suspended ceilings in a laboratory setting. It is particularly relevant for evaluating the acoustic performance of ceiling systems in office and multipurpose buildings. The standard covers aspects like test arrangements, specimen installation, measurement procedures, and CAC calculation, offering a comprehensive framework for assessing and classifying the acoustic properties of ceiling systems under standardized conditions.

ASTM E1425-14

(ASTM E1425-14) focuses on determining the acoustical performance of windows, doors, skylights, and glazed wall systems, testing, and rating their acoustical properties regardless of materials or manufacturing methods. This practice involves testing STL, operating force, latching force, and air leakage, offering a thorough evaluation of these system's acoustical performance while considering various factors that influence sound insulation.

ASTM E2249-19

(ASTM E2249-19) describes the test method for measuring the airborne transmission loss of building partitions using sound intensity, applicable when traditional receiver room requirements cannot be met or to circumvent flanking sound. It offers an alternative to the standard E90 method by directly measuring the sound power radiated by the test element, using Intensity Transmission Loss (ITL) and its corresponding

single-number rating, Intensity Sound Transmission Class (ISTC), to assess the sound insulation properties of building elements in a controlled environment.

ASTM E2611-19

(ASTM E2611-19) is a standard for determining the acoustical properties of porous materials using the transfer matrix method (TMM), involving a setup with four microphones and a digital frequency analysis system. It measures normal incident transmission loss and other critical acoustic properties. It provides a less expensive and time-consuming method than traditional reverberant room methods for evaluating sound absorption and insulation characteristics of porous materials in various applications.

ASTM E2963-22

(ASTM E2963-22) assesses the acoustical effectiveness of ship noise treatments, focusing on bulkhead and deck treatments. These standards measure transmission loss and radiation efficiency and are suitable for evaluating the sound reduction performance of treatments applied to compartment structural partitions like bulkheads, decks, and side shells. It integrates various methods to provide comprehensive insights into the effectiveness of these treatments in reducing airborne noise on ships.

International Organization for Standardization Standards

ISO 16940:2008

(ISO 16940 2008) outlines a method for measuring the loss factor and the equivalent bending rigidity modulus of laminated glass test pieces. Focused on comparing the properties of interlayers used in laminated glass, it details the necessary procedures for these measurements, which are crucial for understanding the STL characteristics of the glazing itself. The standard relates specific parameters, such as the density and thickness of glass components, to the STL, providing a detailed framework for assessing the acoustic properties of laminated glass.

ISO 10140-1:2021

(ISO 10140-1) specifies requirements for laboratory measurement of sound insulation of building elements, including detailed requirements for preparation, mounting of test elements, operating, and test conditions. As part of the ISO 10140 series focusing on acoustics and sound insulation in buildings, this standard provides specific application rules for various building elements to ensure accurate measurement and reporting of sound insulation properties. The standard aims to improve laboratory measurement layouts and ensure consistency across testing scenarios.

ISO 10140-2

(ISO 10140-2) specifies methods for measuring airborne sound insulation of building elements in a laboratory setting, including walls, floors, doors, and windows.

Essential for comparing sound insulation properties of different building materials, it guides the design of elements requiring specific acoustic characteristics. The standard ensures consistency in test environments and methods, providing reliable results crucial for manufacturers and consumers in the construction industry and helping to estimate the in-situ performance of these elements in actual buildings.

ISO 10140-3

(ISO 10140-3) details laboratory methods for measuring the impact sound insulation of floor assemblies, enabling comparison, classification, and design of building elements for specific acoustic properties. It includes guidelines for using a standard tapping machine or alternative sources to simulate different impact noises, such as footsteps or jumping, which are crucial for assessing the sound insulation performance of floors in a controlled, standardized laboratory environment.

ISO 10140-4

(ISO 10140-4) specifies measurement procedures and requirements for laboratory testing of sound insulation of building elements, focusing on basic measurement techniques and processes required for airborne and impact sound insulation testing. It ensures standardized procedures for consistent and accurate measurements applicable to various building elements. It is integral for evaluating sound insulation properties in a controlled laboratory environment and contributes to improved building acoustics and noise control.

ISO 10140-5

(ISO 10140-5) specifies requirements for test facilities and equipment used in laboratory sound insulation measurements in building elements. This standard ensures that laboratories are adequately equipped and designed to perform reliable sound insulation tests, including specifications for laboratory design, test room requirements, and the equipment qualifications used in these measurements, focusing on consistent and accurate testing environments for sound insulation.

Models for Predicting the Sound Transmission Loss in Materials

In the realm of acoustics, understanding how sound travels through different materials is pivotal for designing spaces that meet specific auditory requirements. Navigating the complexities of sound transmission, researchers and engineers rely on various models tailored to address particular aspects of this phenomenon. Table 3 presents various models, each offering a unique perspective on sound transmission through various materials. This exhaustive table encompasses models such as the mass law model, which elucidates the relationship between mass, frequency, and insulation effectiveness based on physical principles of sound transmission. Other detailed models include the spring-mass model, finite element analysis (FEA), and statistical energy analysis (SEA), along with more advanced approaches like machine learning (ML) models and the TMM. Each model is thoroughly explained,

Table 3 A detailed overview of prediction models—This table compares diverse models used to study STL in materials, ranging from traditional principles like the mass law model to advanced techniques like ML models and the TMM

Model	Principle and explanation	Application	Equation	Reference
Mass law model	The mass law model is based on the physical principles of sound transmission through barriers. It posits that an increase in the mass of a partition or the frequency of sound results in a corresponding increase in STL. This relationship is logarithmic and particularly applicable to higher frequencies. The model simplifies the complex nature of sound transmission into a direct relationship between mass, frequency, and insulation effectiveness.	This model is most effective in analyzing the sound insulation properties of single-layer, solid partitions, such as walls in buildings. It is advantageous in contexts requiring critical low-frequency sound insulation, such as residential and commercial buildings. However, its effectiveness significantly diminishes above the material's "critical frequency," where other factors, such as stiffness and internal damping, become more influential.	$\mathbf{TL_{mass}} = 20\log_{10}(f) + 20\log_{10}(\mathrm{m}) - 47.4$ (8) Where f indicates frequency, and m stands for the mass	(Kurtze et al. 1959; Cassidy et al. 2008)
Spring-mass model	The spring-mass model is a dynamic system model that illustrates how complex structures behave under sound transmissions, such as those in double-wall construction or floors with a floating layer. The model incorporates a mass-spring-mass system to simulate the vibrational response of these structures. It accounts for the inertia of the masses (representing rigid layers like walls or floors) and the spring's elasticity (representing insulating	This model is primarily applied in architectural acoustics, particularly for structures that control or mitigate low-frequency noise. Examples include recording studios, concert halls, and residential buildings where sound insulation is crucial. By understanding how sound waves interact with different layers of a structure, architects and engineers can design buildings that effectively minimize sound	$m\ddot{x} + c\dot{x} + kx = F(t)$ (9) Where m is the mass of the object, c is the damping coefficient, k is the spring constant, x is the displacement of the object from its equilibrium position, $\dot{x}$ is the velocity of the object (first derivative of displacement with respect to time), $\ddot{x}$ is the acceleration of the object (second derivative of displacement with respect to time), and $F(t)$ is the external force acting on the system as a function of time t	(Arenas and Ugarte 2016; Yao et al. 2019)

	layers or air gaps). This model is especially adept at demonstrating how sound is transmitted through structures with multiple layers, each exhibiting different vibrational characteristics.	transmission, especially at lower frequencies.		
Finite element analysis	FEA is a computational technique to predict how structures react to external forces, vibrations, and other physical effects. It involves dividing a structure into smaller, finite elements (such as tetrahedrons or hexahedrons) and solving for the behavior of these elements under given conditions. This analysis elucidates how sound, stress, and heat affect a structure. FEA is particularly useful in acoustics for analyzing how sound transmits through various materials and geometries, thereby enabling the design of structures with optimized acoustic properties.	FEA is extensively used in custom designs where precise sound transmission analysis is required. This includes applications such as vehicle cabin design, where minimizing road and engine noise is crucial, and developing complex sound barriers where traditional models prove insufficient. It facilitates tailoring structures to specific acoustic needs, rendering it invaluable in automotive engineering, aerospace, and advanced architectural design.	$[K]u = f$ (10) Where $[K]$ represents the stiffness matrix, which characterizes the stiffness properties of the finite elements and their connectivity within the structure u represents the vector of nodal displacements, which are the unknowns to be solved for. These displacements define the deformation of the structure under the applied loads f represents the vector of nodal forces, which typically includes external loads acting on the structure	(Zienkiewicz et al. 2014; Griese et al. 2015; Papadakis 2018)
Statistical energy analysis	SEA is predominantly employed at high frequencies, where traditional deterministic models become less effective. It conceptualizes structures as collections of interconnected energy systems and employs statistical methods to estimate energy distributions	SEA is commonly used to predict sound transmission in large and complex assemblies, which is paramount in fields such as automotive and aerospace engineering and building acoustics. It is especially apposite for cases like the analysis of the acoustic	$\frac{dE_i}{dt} = \sum_j \left(P_{ji} - P_{ij}\right) dt$ (11) Where $\frac{dE_i}{dt}$ represents the rate of change of energy in subsystem i with respect to time P_{ji} represents the power flow from subsystem j to subsystem i P_{ij} represents the power flow from subsystem i to subsystem j	(Crocker and Price 1969; Lyon and DeJong 1995; Renji 2021)

(continued)

Table 3 (continued)

Model	Principle and explanation	Application	Equation	Reference
	within these systems. SEA is predicated on the principle that, in complex structures, sound energy is distributed among various subsystems and transitions between them in a predictable manner. This method proves particularly effective in environments where sound behavior is complex and diffuse.	properties of aircraft cabins or capacious edifices, where the grandiosity in complexity and scale makes integrating other extrapolative methodologies unfeasible. Through SEA, practitioners can discern how sound energy is distributed throughout these extensive frameworks, thereby identifying critical areas where acoustic fine-tuning can lead to a perceivable uptick in optimality.	The summation $\sum_j$ indicates that the equation sums over all subsystems j within the system	
Empirical models	Empirical models in acoustics are predicated on historical data and employ statistical methods to forecast sound transmission. These models do not rigidly conform to physical laws but depend on discerning trends and patterns within an extensive corpus of accumulated data. Through the analysis of a broad spectrum of historical measurements and outcomes, empirical models are capable of furnishing predictions for sound transmission across a variety of scenarios. Nevertheless, the reliability and precision of these models are contingent upon the	These models prove invaluable for rapid assessments and preliminary design phases in standard constructions, particularly when engaging with conventional materials and configurations. Empirical models hold considerable value in scenarios where expedience is paramount and conditions closely mirror those encapsulated within the historical data set. They find frequent application in the initial stages of building design and the standardization of construction materials and methods.	Based on historical data and statistical methods (no specific equation)	(Flood and Issa 2010; Faruque 2015; Özen et al. 2019; Mian et al. 2022)

	quality and pertinence of the underlying data. Consequently, their accuracy is circumscribed to conditions and materials analogous to those represented in the dataset.			
Machine learning models	ML models in acoustics employ algorithms to identify patterns and make predictions based on extensive datasets. These models "learn" from data, improving their predictions over time. ML is particularly effective for analyzing materials and structures characterized by complex, non-linear acoustic properties. Capable of adapting to new information, these models are highly versatile and adept at managing various acoustic prediction tasks.	These models are increasingly valuable as material databases expand, particularly in fields where traditional models fall short. Their applications encompass predicting the STL of novel materials, optimizing the acoustic design of complex structures, and even in noise control engineering. The quality of predictions is highly dependent on the quality and quantity of the training data, making the collection of comprehensive and diverse data sets crucial.	Utilizes ML algorithms to identify patterns (no specific equation)	(Umehara et al. 2019; McCarthy et al. 2023)
Wave-based models	Wave-based models are essential in acoustics for accurately representing the wave nature of sound. These models account for various wave phenomena, such as diffraction, absorption, and interference. They are founded on solving differential equations that describe sound wave propagation and interaction within different environments. These models are particularly adept at capturing the	These models are crucial in the design of acoustically sensitive environments, such as concert halls, recording studios, and theaters. They facilitate a detailed understanding of how sound waves interact with the environment, enabling the creation of spaces with optimal acoustical properties. This is particularly important in environments where sound quality and clarity are paramount.	It involves solving differential equations for sound wave propagation (specific equations vary based on the model)	(Candy and Sullivan 1996; Lam 2005; Dijckmans and Vermeir 2012)

(continued)

Table 3 (continued)

Model	Principle and explanation	Application	Equation	Reference
	complexities of sound behavior in various mediums and are crucial for designing spaces with specific acoustic requirements.			
Transfer matrix method	TMM represents a sophisticated analytical tool for evaluating sound transmission through multilayered structures. It entails calculating the transfer matrix for each layer within a structure and computing the overall sound transmission. This method facilitates a comprehensive analysis of how sound interacts with each material interface within a multilayered system, offering detailed insights into the acoustic properties of complex structures.	TMM is particularly effective in analyzing structures such as laminated glass or composite walls, where the material properties of each layer significantly impact the overall sound transmission. It is extensively used in the construction and automotive industries to design materials and structures that effectively manage sound transmission. This makes it invaluable for creating environments that are both quiet and comfortable.	$\begin{bmatrix} P_1 \\ U_2 \end{bmatrix} = [T] \begin{bmatrix} P_2 \\ U_2 \end{bmatrix}$ (12) Where P_1 and P_2 represent the acoustic pressures at the input and output sides of the multilayered structure, respectively U_1 and U_2 represent the particle velocities at the input and output sides of the multilayered structure, respectively $[T]$ Is the transfer matrix, which characterizes the sound transmission properties of the entire multilayered structure	(Bolton et al. 1996; Song and Bolton 2000; Song et al. 2023)

detailing its underlying principle, scope of application, and the relevant equations. This provides a comprehensive perspective on acoustic modeling across different contexts.

Commercially Utilized Sound Insulation Materials

In our comprehensive guide on commercially utilized sound insulation materials, Table 4 not only lists but also compares various sound insulation options based on key parameters. This table offers a broad perspective on materials like fiberglass insulation, acoustic foam panels, and mass-loaded vinyl, among others, highlighting their STC ratings, effective frequency ranges, and approximate costs per square

Table 4 Commercially utilized sound insulation materials—A detailed view of the STC ratings, frequency ranges, and costs per square meter, facilitating informed decision-making for soundproofing projects

Sound insulation material	Sound transmission class (STC)	Frequency range	Cost per square meter (Approx.)	Source of information
Fiberglass insulation	25–45	Low to mid-range	$10–$30	(Insulation4US)
Acoustic foam panels	20–35	Mid to high-range	$15–$50	(Arrowzoom Acoustics)
Mass-loaded vinyl (MLV)	20–30	Low to mid-range	$20–$40	(Acoustical surfaces)
Green glue	50+	Mid to high-range	$30–$60	(SoundAway)
Acoustic mineral wool	30–60	Low to mid-range	$10–$30	(Acoustimac)
Soundproof drywall	45–70	Low to mid-range	$20–$40	(Amazon.com)
Acoustic plywood	25–50	Low to mid-range	$20–$40	(Acoustimac)
Acoustic ceiling tiles	Varies	Mid-range	$10–$30	(Amazon.com)
Double glazed windows	35–50	Mid to high-range	$100–$300	(Second skin audio)
Acoustic curtains	15–25	Mid-range	$20–$60	(Saaria)
Soundproof wallpaper	Varies	Mid-range	$10–$30	(Selected wallpapers and interiors)
Acoustic insulation blankets	20–30	Mid-range	$10–$30	(Acoustical surfaces)
Soundproof doors	30–50	Low to mid-range	$100–$500	(Vigor doors)

meter. It also includes advanced options like green glue and acoustic mineral wool, showcasing their superior sound insulation properties. For those seeking specific solutions, the table provides insights into specialized products like acoustic ceiling tiles, double-glazed windows, and soundproof doors, ensuring a wide range of choices for different soundproofing needs.

Recent Advances in Sound Insulation Materials

Metamaterials

Metamaterials, known for their extraordinary ability to manipulate sound wave propagation, have sparked significant interest in acoustic engineering due to their unparalleled STL capabilities. These materials have been the focus of extensive research, with studies demonstrating their potential to revolutionize sound insulation, especially at low frequencies.

Metamaterial-based thin plates equipped with subwavelength resonator arrays exhibit higher STL than conventional bare plates, showcasing their effectiveness in low-frequency sound insulation (Xiao et al. 2012). A thick metamaterial plate with spring-mass resonators can achieve STL up to 72 dB at 2 kHz, significantly surpassing the performance of homogeneous plates (Oudich et al. 2014).

Membrane-type acoustic metamaterials provide excellent sound attenuation in the low-frequency range, with STL being strongly influenced by the attached mass and membrane properties (Zhang et al. 2012). The critical role of damping within metamaterials for vibroacoustic attenuation performance, particularly the impact of damping in resonators and host structures on STL, is highlighted (Van Belle and Desmet 2017).

Lightweight multilayer honeycomb membrane-type acoustic metamaterials (Fig. 5a) defy the traditional mass law to achieve exceptional STL while maintaining a low weight (Li et al. 2020). Acoustic metamaterials incorporating piezoelectric patches offer dual functionality in sound insulation and energy harvesting, achieving over 20 dB STL and a maximum energy conversion efficiency of up to 15.3% (Li et al. 2016).

Helmholtz cavity-based acoustic metamaterials demonstrate exceptional sound insulation performance, particularly in isolating low-frequency noise (He et al. 2019). Metamaterial barriers based on cellular structures (Fig. 5b) provide high STL in the low-frequency range without the addition of excessive mass (Varanasi et al. 2017). The significance of damping in locally resonant metamaterial plates (Fig. 5c) is noted, with the damping design of the resonators affecting the acoustic insulation performance (Van Belle et al. 2019).

Coupled-resonance acoustic metamaterials promise broadband and intense STL, offering practical configurations for adjusting frequency ranges and facilitating mass fabrication (Roca and Hussein 2021). Metamaterials designed with meta-porous materials (Fig. 5d) tuned at exceptional points optimize sound attenuation in acoustic ducts at low frequencies (Xiong et al. 2017).

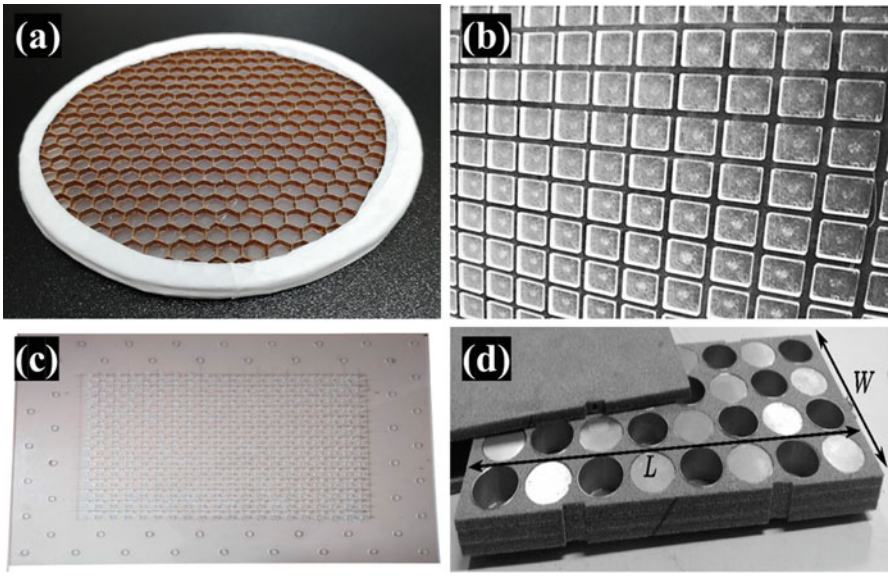


Fig. 5 (**a**) Multilayer honeycomb membrane (Li et al. 2020). (With the permission of Elsevier © [Elsevier]), (**b**) Cellular structure (Varanasi et al. 2017) (With the permission of AIP Publishing © [AIP Publishing]), (**c**) Locally resonant metamaterial plates (Van Belle et al. 2019) (With the permission of Elsevier © [Elsevier]), (**d**) Meta-porous material (Xiong et al. 2017) (With the permission of AIP Publishing © [AIP Publishing])

In summary, the extensive research on metamaterials underscores their significant advancements in STL, particularly at low frequencies. These materials offer promising applications in building acoustics, aeronautical noise control, and beyond, outperforming traditional materials in sound insulation while maintaining low mass and thickness, paving the way for future innovations in the field.

Perforated Panels

Perforated panels are widely studied for their STL properties, offering significant implications for acoustic engineering and noise control. A study on thick perforated panels found that perforations reduce the coincidence effect at critical frequencies, enhancing STL above 630 Hz (Chen and Jan 2001). The hole diameter significantly affects STL, with smaller diameters in micro-perforated panels resulting in higher STL due to resistive parts in the hole impedance (Putra and Ismail 2014).

A fully coupled analysis of fluid-loaded finite perforated panels set in a rigid baffle shows that most transmission occurs through the perforations, particularly in light mediums like air (Mana and Sonti 2016). The acoustic behavior of 3D-printed bio-degradable micro-perforated panels (Fig. 6) with varying perforation cross-sections indicates better sound absorption than uniform cross-sectional perforation

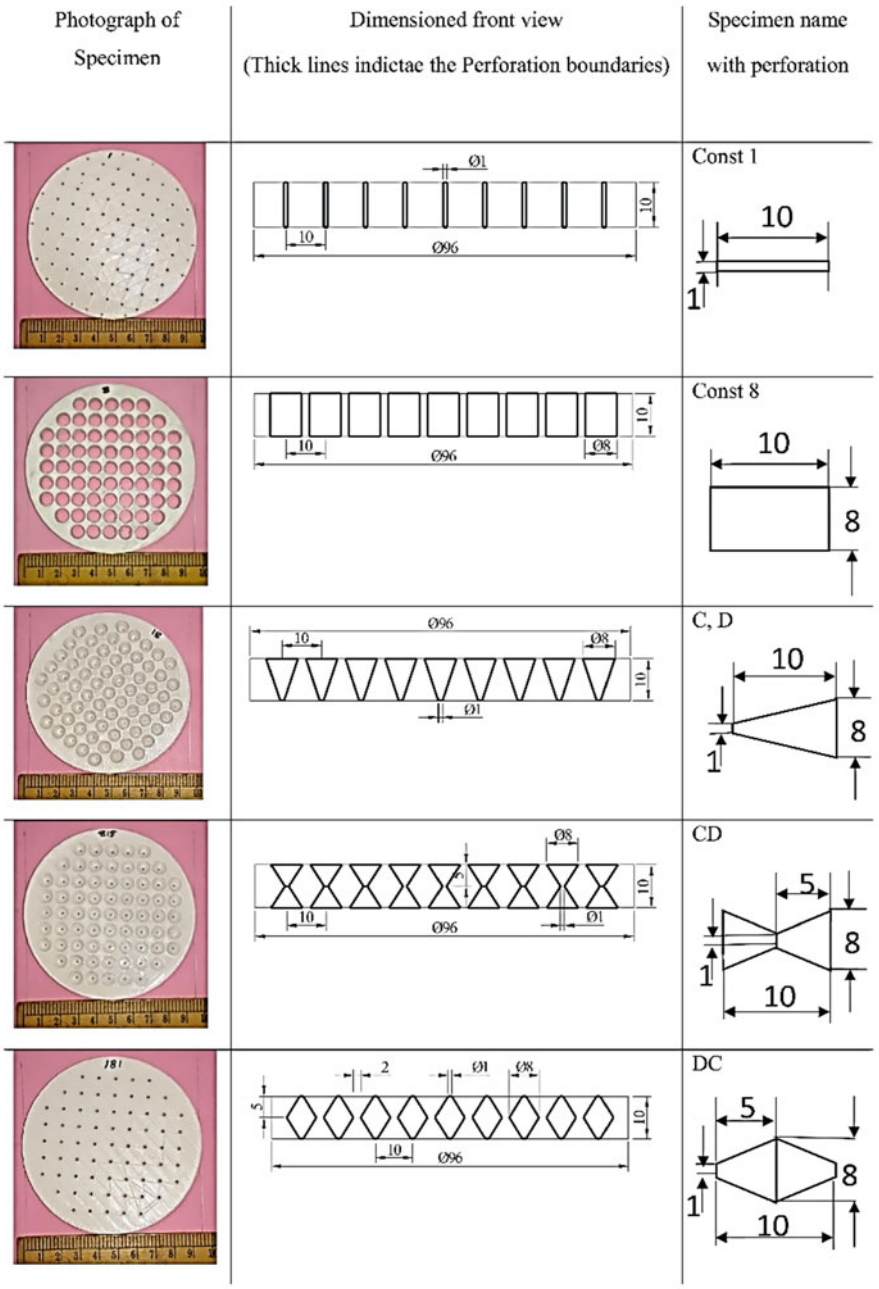


Fig. 6 3D printed specimens with details of perforation cross-section variations. (All dimensions are in mm) (Sailesh et al. 2021). (With the permission of Elsevier © [Elsevier])

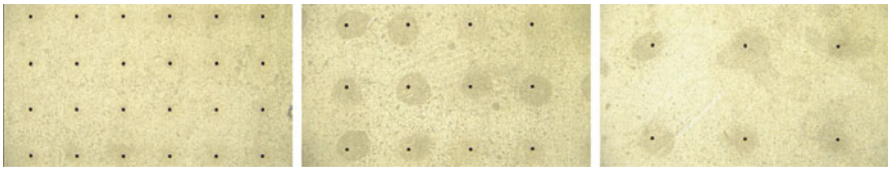


Fig. 7 Pictures of the sample with different perforation rates (Hou et al. 2021). (With the permission of Elsevier © [Elsevier])

(Sailesh et al. 2021). The STL of perforated-natural fiber sandwich panels is notably influenced by perforation diameter, with a hybrid panel combining pineapple leaf fiber and a 3 mm perforation achieving the highest STL among tested samples (Hong Tan and Hooi Ak 2020).

Small perforations in corrugated sandwich panels significantly enhance low-frequency sound absorption and transmission loss (Meng et al. 2017). Underwater flexible micro-perforated panels (Fig. 7) show that the panel's perforation rate and elastic properties significantly influence STL (Hou et al. 2021). Micro-perforated panels coupled with honeycomb panels consider air cavities for offshore plants, showing a significant increase in STL at mid-frequencies (Jung et al. 2019). Additionally, the effect of perforating honeycomb walls of a honeycomb sandwich panel on dynamic bending stiffness and STL highlights the significance of the perforation design (Kang et al. 2022).

These studies underscore the critical role of perforation characteristics in enhancing the acoustic performance of perforated panels, specifically in terms of STL, across a range of applications. Exploring different materials, perforation sizes, and configurations opens avenues for further research and innovation in noise control and acoustic optimization.

Polymer Composites

STL is critical in the functionality optimization of polymer composites for various applications, such as noise reduction and insulation. The research underscores that the acoustic properties of these materials are significantly influenced by their physical composition and structure.

The sound insulation behavior of inorganic particulate-filled polymer composites has been extensively studied, revealing that STL increases with the glass bead volume fraction and sound frequency. However, it decreases when the size of the glass beads is increased, given a constant volume fraction (Liang and Jiang 2012). This trend is consistent across various composite types, including those filled with hollow microspheres (Liang and Zhu 2016). The addition of carbon nanotubes (CNTs) into Acrylonitrile Butadiene Styrene (ABS) composites has been shown to significantly enhance STL, a benefit attributed to the increased stiffness provided by the CNT additives (Lee et al. 2008).

Further investigations into polymer/inorganic particle composites, such as PP/nano-$CaCO_3$, ABS/$CaCO_3$, and PVC/hollow glass bead (HGB), have confirmed the significant role of particle type and composite structure on STL (Liang 2015). Moreover, adding CNTs and foaming to wood fiber-low-density polyethylene (LDPE) composites has improved STL, especially at medium and higher frequencies (Neyciyani et al. 2017).

Hybrid composites, particularly those reinforced with flax fabric and polypropylene (PP) hybridized with carbon fibers (Fig. 8), have demonstrated STL per areal density significantly higher than traditional materials like aluminum, with further improvements upon hybridization (Haris et al. 2019). Benchtop laboratory sound testing systems have shown potential as a cost-effective method for evaluating the STL of polymer and polymer composite samples (Allan et al. 2012). Optimizing carbon fiber-reinforced polymer (CFRP) composites can achieve transmission loss levels comparable to traditional metal materials, especially at lower frequencies (Zhang and Du 2017).

Synergistic effects on STL have been observed in PP/clay/CNT nanocomposites, with the STL for these composites significantly outperforming pure PP at both high and low frequencies (Kim et al. 2013). Research on composite rubber reinforced with hollow glass microspheres (HGM) indicates promising noise reduction capabilities and excellent sound insulation performance, with optimal weighted STL achieving significant levels under various thickness constraints (Yang et al. 2023). Furthermore, enhancing polymer nanocomposites with nano-clay particles and nanofibers in polyurethane foam can increase STL and the absorption coefficient (Park et al. 2019).

In summary, the extensive research on STL in polymer composites highlights diverse strategies for enhancing their acoustic properties through innovative material composition and structural modifications. These studies advance materials science and pave the way for more efficient and effective sound insulation solutions.

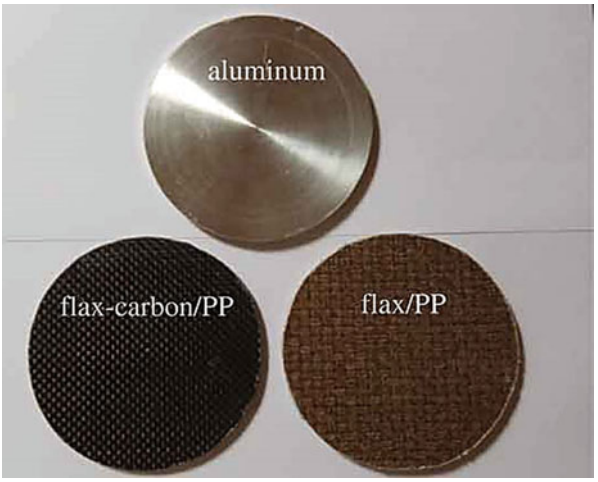


Fig. 8 PP-based composites reinforced with a hybrid structure of flax fabric and carbon fabric (Haris et al. 2019). (With the permission of Taylor & Francis © [Taylor & Franics])

Recycled or Waste-Derived Sound Insulation Materials

Recent research in sound insulation materials has yielded significant advancements, with studies exploring various recycled and natural materials for soundproofing applications. These efforts aim to identify sustainable solutions that offer effective noise reduction in various applications. (Zhao et al. 2010) studied the sound insulation properties of wood-waste tire rubber composite panels (WRCP). They found that these panels exhibit superior soundproofing compared to commercial wood particleboard and composite floorboard. Key findings include the observation that increasing the rubber crumb content and the level of PMDI adhesive in the WRCP enhances its sound insulation properties. Subsequently, (Hung et al. 2014) focused on the sound insulation properties of inorganic polymeric foams (IPF) created from metakaolin and blast furnace slag. They observed significant STL improvements in IPFs with optimal densities, demonstrating their potential as sound-insulating materials in construction and building applications. The study by (Lee and Kim 2014) investigates the sound absorption and transmission loss performances of natural and processed woods, including cork, when combined with various natural or waste materials. It found that natural and hard-processed woods were generally poor sound absorbers but effective sound barriers, with maximum transmission loss up to approximately 70 dB. Adding materials like coconut fiber filter mats significantly improved sound absorption in hard-processed wood. Cork, classified as a soft processed wood, showed better sound absorption at mid-frequency ranges (500 Hz to 2000 Hz) and had acceptable transmission loss of up to 40 dB. The study recommends cork for applications requiring both sound absorption and transmission loss, highlighting its effectiveness when combined with materials like corn fiber tea bags.

(Gandoman and Kokabi 2015) investigated the sound barrier properties of sustainable waste rubber/geopolymer concretes. They developed a novel concrete using metakaolin-geopolymer with waste rubber, aiming for enhanced noise barrier applications. The study reveals that waste rubber geopolymer concrete (WRGC) exhibits superior STL and better sound absorption properties than conventional concrete. It demonstrates increased NRCs, indicating its effectiveness as a sound barrier material. (Naghmouchi et al. 2016) focused on the soundproofing characteristics of olive stone-filled PP, gypsum boards and wood fiber-reinforced PP. The study involved preparing composite materials using olive stone flour and examining their soundproofing properties through impedance tube testing. The results were compared with those for gypsum boards, wood veneer, and wood fiber-reinforced PP, providing insights into the acoustic performance of these materials. (Ferrández-García et al. 2017) examined the acoustic and thermal properties of palm panels made from Washingtonia palm tree pruning waste. Three boards with particle sizes ranging from 0.25 to 4.00 mm were created using 8% urea-formaldehyde resin, showing superior mechanical performance compared to traditional insulating boards. The average thermal conductivity was consistent at 0.062 W/(mK) across all sizes. The panels achieved a class D acoustic rating at low frequencies (125 and 250 Hz).



Fig. 9 WPCB-UP composites (Cai et al. 2017). (With the permission of Elsevier © [Elsevier])

(Cai et al. 2017) explored the sound insulation properties of composites made from waste-printed circuit boards (WPCBs) and unsaturated polyester (UP) (Fig. 9). They found that smaller WPCB particle sizes and higher WPCB mass ratios in the composite enhance surface density and stiffness, contributing to better sound insulation. The study demonstrates that these composite's maximum weighted sound reduction index can reach up to 28.4 dB, suggesting their potential as efficient soundproofing materials. In the paper by (António et al. 2018), rice husk waste is combined with expanded cork and recycled rubber granules to develop new composite boards for construction. The study demonstrates these composites' thermal and acoustic insulation potential, with weighted reductions in impact SPLs ranging from 20 to 27 dB. The thermal conductivity of the composites varies between 0.060 and 0.074 W/(m·°C), suggesting their effectiveness in building insulation. The study by (Can 2019) focuses on developing sound insulation materials using recycled ABS reinforced with short cotton fiber waste. The research reveals that these composites exhibit substantial STL, with the highest mean value reaching around 70 dB.

(Cole and Street 2019) examined the impact of adding varying amounts of micronized rubber powder (MRP) to pine particleboard on its acoustic and mechanical properties. The study found that increasing MRP content decreased the strength and increased the flexibility of the particleboards. Acoustically, MRP addition slightly improved STL but did not significantly enhance sound absorption compared to the control sample. This suggests MRP's potential for specific applications where reduced strength is acceptable and enhanced flexibility or slight improvements in sound insulation are desired. The research conducted by (Barbanera et al. 2020) focuses on using recycled leather cutting waste (LCW) to produce acoustic panels (Fig. 10). The study explores these panels' sound absorption and transmission loss properties. The results indicate that the LCW-based panels exhibit significant sound absorption capabilities, with absorption coefficients ranging from 0.2 to 0.7 in the mid-frequency range. The STL values are up to 30 dB, demonstrating their effectiveness in sound insulation. The study by (Sezgin et al. 2021) focuses on the design of composite insulation panels made entirely from recycled cotton fibers and polyethylene/PP packaging wastes. These materials were processed into panels and tested for thermal and acoustic insulation properties. The panels demonstrated



Fig. 10 LCW-based samples for acoustical testing (Barbanera et al. 2020). (With the permission of Springer Nature © Springer Nature)

notable STL, reaching 8 dB, and improved thermal resistance, with a value of up to 0.11 $m^{2\circ}C/W$.

Nastac et al. (Nastac et al. 2021) delved into developing composite materials using expanded perlite, natural polymers, and rapeseed waste for sound insulation. The study showcases significant improvements in sound absorption coefficients at various frequencies, highlighting the composites' potential for noise reduction. (Mehta et al. 2021) explored recycling solid high-energy propellant (HEP) waste into fired clay bricks, focusing on enhancing their acoustic properties. The study experimented with bricks containing different percentages of HEP waste, revealing that adding HEP waste affects the STL depending on the orientation and proportion of waste. Specifically, bricks with HEP waste exhibited changes in STL, with directional orientation during baking influencing acoustic performance. This innovative approach addresses waste management challenges and contributes to developing construction materials with improved acoustic insulation properties. (El Messiry and Ayman 2022) research investigates the STL of composite panels made from natural fibers and rubber crumbs. Their study demonstrates that panels made from a combination of fiber (cotton wool and Kapok) and rubber crumbs exhibit significant STL profiles, with Kapok panels achieving up to 47 dB. Adding polylactic acid melt-blown nonwoven fabric on the rubber crumbs side increases the STL by approximately 20%.

The study by (Islam et al. 2022) investigates using post-consumer recycled denim fabrics to develop acoustic insulation materials. They found that incorporating Sorona® or PLA binder fibers with recycled denim fibers enhances the material's sound insulation properties. The study demonstrates that at around 1000 Hz, these materials achieve a maximum transmission loss of approximately 24 dB, and the transmission coefficient is close to zero. The study by (Dönmez and Turker 2022) focuses on producing building panels using recycled waste fibers from yarn

factories. The panels are tested for their thermal and sound insulation performances. The findings reveal that the thermal conductivity coefficient values of the produced composite panels range between 0.033 and 0.038 W/mK. Additionally, the STL values range between 24.2 and 32.6 dB, indicating their effectiveness in insulation. (Sunali et al. 2022; Jaish et al. 2023a) study on developing natural rubber-bamboo biochar (NR-BB) composites for vibration and noise control applications marks a pioneering effort to harness bamboo biochar as a filler to enhance NR composites' acoustical and vibration-damping properties. The study identifies significant improvements in the vibration damping ratio or system loss factor, achieving a maximum ξ value of 33.19 for NR alone, which decreases to 17.46 and 13.50 for NR/10BB and NR/20BB composites, respectively, in the free layer damping configurations. STL shows notable enhancements, with an 8% and 11% increase for the 10 and 20 phr BB loadings compared to the NR composites. The research underscores the acoustical and vibration-damping advancements and highlights the positive impact of BB loading on the NR composites' physical, mechanical, and thermal properties.

Açıkgöz et al. (2023) explore the development of acoustic panels from post-production curtain waste, specifically blackout and dimout fabrics. Utilizing a hot press method, the research produced panels with varying thicknesses and compositions, analyzing their physical and acoustic properties. Key findings include enhanced sound absorption and transmission loss with increased thickness and the blend of blackout and dimout materials, demonstrating the potential of recycled curtain waste in creating sustainable, decorative, and efficient acoustic insulation solutions. (Mago et al. 2023) study investigates the sound insulation properties of recycled high-density polyethylene (rHDPE) combined with waste jute fabric (WJF) composites developed through compression molding. Incorporating a 10% by-weight WJF layer into the rHDPE matrix significantly enhances tensile strength and modulus by approximately 18.24% and 46.52%, respectively, while improving STL performance. The research underscores the potential of using recycled and waste materials in developing efficient sound-insulating composites, offering a sustainable alternative for acoustic material design. (Astrauskas et al. 2023) investigate the acoustic properties of structures made from perforated plastic panels (PPP) backed with rubber granule panels (RGP), utilizing 3D printing and recycled materials. The research focuses on sound absorption and transmission loss, demonstrating the impact of perforation angles and RGP thickness on acoustic performance. Through multi-criteria analysis (TOPSIS), the study identifies the most effective configurations for sound absorption and insulation, contributing to sustainable material use in acoustical applications. (Jaish et al. 2023b; Sunali et al. 2024) research focuses on enhancing NR composites' thermal, vibrational, and acoustical properties by incorporating pyrolytic carbon black (PCB) derived from tire waste. Increasing PCB content to 30 phr in the NR matrix significantly improves thermal conductivity by approximately 21.16% at 40 °C and enhances damping ratios, with the highest damping ratio of 6.56 observed at the third vibration mode. Additionally, the study demonstrates a notable increase in STL of about 17.19% with the

NR/PCB30 composite compared to NR/PCB0, highlighting PCB's effectiveness in improving the desired properties of NR-based composites.

In conclusion, sound insulation materials have grown considerably, with research from natural substances to innovative composites of recycled and waste materials. These studies collectively underscore the potential for achieving superior soundproofing through sustainable means, marking an exciting trajectory for future research and application in the industry.

Practical Applications of Sound Insulation Materials

Sound insulation materials are widely used in various fields to control, dampen, or block noise. Their practical applications are diverse and essential in many industries and environments. Here are some of the critical applications (Ko et al. 2007; Wester et al. 2009; Maldovan 2013; Zhou et al. 2014; Sung and Kim 2017; Fei et al. 2018; Gulia and Gupta 2018; Haliq et al. 2019; Deuszkiewicz 2021; Sunali et al. 2022, 2024; Ghassabi and Talebitooti 2023; Kinetics noise control 2024):

i. **Construction and Architecture**: Sound insulation materials are crucial in building construction to reduce noise transmission through walls, floors, and ceilings. This is particularly important in multi-unit residential buildings, hotels, schools, hospitals, and office buildings, where noise reduction contributes significantly to comfort and privacy.
ii. **Automotive Industry**: In vehicles, sound insulation is used to reduce the noise of the engine, road, and wind. This improves the comfort level inside the vehicle, making for a quieter and more pleasant driving experience.
iii. **Manufacturing and Industrial Settings**: Factories and industrial plants often use sound insulation materials to reduce machinery noise. This is not only for the comfort of workers but also to comply with occupational health and safety standards regarding noise levels.
iv. **Home Appliances**: Sound insulation is used in home appliances like dishwashers, washing machines, and refrigerators to minimize operational noise, making them more user-friendly in residential settings.
v. **HVAC Systems**: Heating, ventilation, and air conditioning systems can be significant noise sources in buildings. Sound insulation materials minimize the noise these systems produce, enhancing the comfort of indoor environments.
vi. **Transportation Infrastructure**: Sound barriers along highways and railways are made from sound insulation materials to protect residential areas from traffic noise. These barriers help reduce the impact of transportation noise on nearby communities.
vii. **Specialized Use in Sensitive Environments**: In environments like libraries, hospitals, and schools, where quietness is essential, sound insulation materials help create a conducive atmosphere for study, rest, or recuperation.

viii. **Audio Equipment**: High-quality speakers and audio equipment often incorporate sound insulation materials to improve sound quality by minimizing distortion and interference.
ix. **Personal Protective Equipment (PPE)**: In industries with high noise levels, workers use PPE like earmuffs and earplugs made from sound insulation materials to protect their hearing.

Future Directions and Possibilities

The field of acoustics and noise control is poised for a future characterized by continuous evolution, driven by the convergence of technological advancements and the pressing demands of society. In this section, exploration focuses on potential trends and developments proposed to shape sound insulation's future underpinned by scientific understanding.

Advancements in Acoustic Materials

One of the most prominent areas of exploration lies in the ongoing research and development of novel acoustic materials with the potential to revolutionize sound insulation. These materials are envisioned to not only enhance sound-blocking properties but also to embrace principles of sustainability, cost-effectiveness, and versatility. Breakthroughs in materials science are expected to usher in innovative solutions across diverse industries, offering superior noise reduction and acoustic performance (Han et al. 2024; Cho et al. 2024; Qin et al. 2024).

Smart Acoustic Technologies

Integrating intelligent technologies into sound insulation systems stands out as a compelling avenue for the future. This involves using advanced sensors, actuators, and artificial intelligence algorithms to create adaptive acoustic environments. These smart systems are envisioned to offer real-time noise control and personalized comfort, adapting dynamically to ever-changing acoustic conditions (Kusiak 2018).

Sustainable and Green Sound Insulation

Exploring eco-friendly sound insulation practices is imperative in an era where sustainability takes center stage in construction and design. This involves evaluating how sound insulation can align with environmentally conscious principles. Key considerations encompass using recycled materials, adopting energy-efficient designs, and implementing environmentally friendly manufacturing processes, all

aimed at minimizing the ecological footprint of sound insulation solutions (Arenas and Sakagami 2020).

Evolving Urban Landscapes

The ongoing trend of urbanization poses formidable challenges related to noise pollution in densely populated urban areas. It is incumbent upon experts to explore the intricate interplay between sound insulation and the ever-evolving urban landscapes. The focus is on understanding how urban planning, zoning regulations, and architectural innovations can mitigate noise and enhance the overall quality of life in cities. Scientifically sound urban acoustic design will be instrumental in this endeavor (Paskaleva 2011).

Regulatory Changes and Standards

Anticipating potential changes in noise regulations and standards at the local, national, and international levels is paramount. A deep dive into the dynamic nature of regulatory frameworks governing noise control is necessary, along with exploring their potential impacts on sound insulation requirements across diverse industries and sectors. Scientific insights will be crucial in assessing how evolving standards may shape the design and implementation of sound insulation systems in the future (Ferraris et al. 2018).

Interdisciplinary Collaborations

The synergy from collaborative efforts across diverse disciplines holds immense promise in effectively addressing the multifaceted challenges of acoustics and noise control. Exploring the potential of interdisciplinary collaborations, encompassing fields such as architecture, engineering, psychology, and healthcare, in great detail is essential. Scientifically grounded approaches integrating expertise from multiple domains are expected to yield innovative solutions to noise-related issues, emphasizing evidence-based practices (Tao et al. 2021).

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